

Evaluation of an interaction design for haptic directional guidance.

Pedro Gomez Hernandez*

Business Development and Technology dep., Aarhus University, 7400 Herning, DK

ABSTRACT

This study investigates the design of a haptic feedback interaction in a guidance task using a wearable system that provides fingertip force feedback. The user is guided towards an invisible target by using directional cutaneous cues. The aim is to find if introducing force variation in this feedback interaction would result in a better performance than a force static sensation. Despite no significant differences in task completion time or accuracy has been found, trends indicate that static feedback was rated with a higher realism score compared to the varying feedback but with slower completion times.

Index Terms: Haptics, Design, Cutaneous Feedback

1 INTRODUCTION

In the realm of human-computer interaction and task performance, feedback plays a crucial role in guiding users toward desired outcomes. Feedback can be presented in various forms, such as visual, auditory, or haptic, and its design could influence user's task completion performance results (e.g. speed, accuracy). In virtual reality scenarios for example, the utilization of haptic feedback has been shown to result in more accurate performance in guidance tasks [7] [3] [12].

As haptic devices advance and become more compact, these devices come with some trade-offs: arm-workspace sized devices like Haption-Virtuous allow for realistic kinesthetic feedback sensation, with the drawback of constraining the user workspace. Wearable haptic devices with cutaneous actuators, allow to the user have freedom of movement within a confined space, yet the haptic sensation they provides could be limited to only skin, thus not so realistic.

This paper proposes an experimental scenario to evaluate the design of a feedback interaction that aims to create a more intuitive sense of directional "push". While addressing the challenges caused by utilizing static cutaneous feedback alone. An early pilot study was conducted to analyze how variations in cutaneous force feedback influence user perception and guidance.

The investigation focuses on understanding the effect of different variations within the same feedback modality, in order to evaluate the task performance, in this experiment the position accuracy and completion time have been considered as metrics. Insights gained from this study could advance the design of haptic interactions which need established principles, to help create effective tactile experiences [2], research shows that understanding user behavior and feedback mechanisms help improve haptic interactions [16].

Through an exploration of these relationships, the findings aim to enrich the broader understanding of interaction feedback design and its impact on performance. The aim is to offer knowledge for designing more effective feedback interaction systems in human-computer interaction, enhancing both theoretical frameworks and user interface design.

*e-mail: pedrogh@btech.au.dk

2 RELATED WORKS

2.1 Haptic guidance

The haptic guidance leverages tactile feedback to enhance performance in diverse applications such as surgical training, navigation, and teleoperation tasks. By integrating sensory feedback, haptic guidance improves task accuracy, reduces cognitive load, and enhances user experience. For instance, in robot-assisted surgical training, trajectory prediction algorithms provide resistive and assistive cues to improve operator performance and post-training outcomes [13]. In teleoperation, haptic feedback enhances control and efficiency in tasks like obstacle avoidance and precision drilling, reducing completion time and physical effort [4] [9].

2.2 How cutaneous feedback is generated

Wearable haptic feedback systems provide the sensation of pressure through fingertip actuators by applying controlled forces that mimic real-world interactions, such as pressing against a surface or resist to a weight [11] [15].

The realism of this sensation depends on factors like force magnitude, direction, and temporal characteristics. Accurate force feedback, aligned with visual and proprioceptive cues, is crucial for users to perceive the interaction as natural and realistic [6] [5].

The design of this system aims to provide a user with the feeling that a fingertip would experience when presented with directional force perpendicular to the fingertip.

2.3 Mechanoreceptor saturation

Fingertip mechanoreceptor saturation happens when the sensory receptors in the skin, reach their maximum response and can no longer detect further changes. This affects touch perception, especially when pressure or vibration is constant. Some receptors are sensitive to high-frequency vibrations but become less responsive if exposed to them for too long [10]. The amount of pressure applied and how much the skin is pushed or indented also influence how these receptors react [10] [8].

Reaching fingertip mechanoreceptor saturation could result in altering user's perception, affecting the users' ability during the task. This paper explores different patterns modulating the force received, and compares them to static force to see if the system reaches mechanoreceptors saturation and if the modulations avoid that issue.

3 METHODOLOGY

This study uses mixed-methods to evaluate the proposed interaction method, the methodology was designed to assess both the performance of the designed feedback interaction using objective metrics (completion time and accuracy) and the subjective experience of users with a set of questions targeting perceived realism and intuitiveness. For the quantitative metrics a specific task has been designed. The task consists of a target finding task utilizing haptic guidance to determine the differences between feedback variations

Quantitative Evaluation: Objective metrics, including task completion time and the user's hand positional accuracy were collected to evaluate user performance. The users positional accuracy was determined by measuring the users hand vertical position (blue sphere in Fig. 1) to an series of invisible targets present on the vertical line in Fig. 1. Completion time was measured as the time for

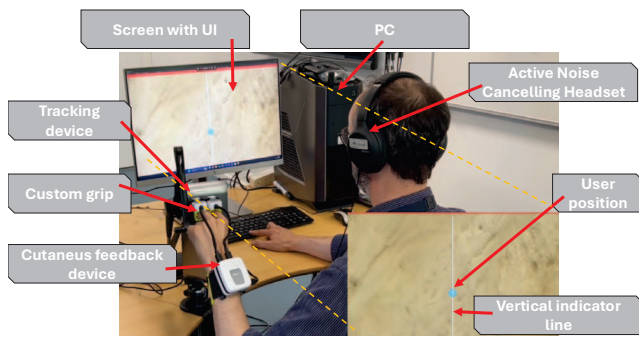


Figure 1: Experimental setup and user interface.

the user to reach each target from a middle point standardized initial position at the middle of the line. The complete task consists of reaching 18 different targets, that were divided into 3 groups and the total travel distance for each group of 6 targets was designed to be constant, as an example on how the constant travel distance works, one set could include the user's hand travel distances of 7 cm, 5 cm, 8 cm, 6 cm, 4 cm, and 10 cm, adding up to 40 cm. Another possible set might feature measurements of 6 cm, 7 cm, 6 cm, 5 cm, 8 cm, and 8 cm, also totaling 40 cm; so the total travel distance remains the same. These metrics offer a set of measures for assessing the system's effectiveness and facilitate comparisons with similar interaction techniques. By quantifying both the time required to reach the desired position and the deviation from the target, these metrics serve as critical benchmarks widely utilized in the field for evaluating interaction systems[12] [1] [14].

Qualitative Evaluation: User feedback was gathered through a questionnaire to assess the subjective experience of the feedback during the task. Participants were asked whether the feedback effectively simulated a "push" sensation perceived as the feeling of force or pressure being applied to the user's fingertips to mimic the action of being physically pushed. This qualitative evaluation aimed to determine the extent to which the system conveyed the intended haptic feedback.

4 PROPOSED INTERACTION DESIGN

To provide the user with a sensation of a directional push feedback using the wearable device, the concept involves replicating the force sensations that would naturally occur on the fingertip if the user were physically holding an object and that object was in motion. These forces typically include normal (perpendicular to the skin surface) and shear forces (parallel to the skin surface), which combine to produce the sensation of direction and intensity of movement. This interaction design is tested on a device that can reproduce perpendicular force to the fingertip. With this limitation, the forces exerted on the fingertips surface can only suggest a user to move along a single direction. Theories from haptic perception, such as the mechanoreceptors response model, explain how the skin's mechanoreceptors detect and interpret these forces to convey tactile feedback. By simulating these forces through actuators or similar mechanisms in the device, the system can recreate the feeling of an object's motion, effectively tricking the brain into perceiving dynamic interactions with virtual objects, providing the user with the desired sense of direction.

In order to enable the generation of forces required for the task, the WEART TouchDiver was utilized. This device is capable of providing vertical forces, texture representation, and temperature variation on the fingertip surface, the device achieves this by utilizing a mobile flat surface that is in contact with the fingertip containing a Peltier module that provides the temperature variations.

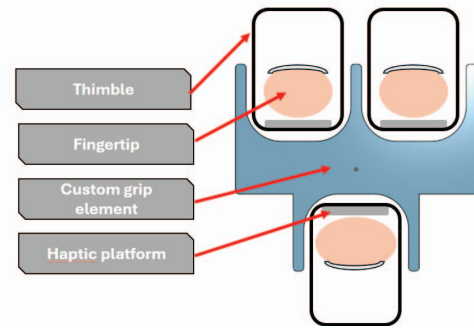


Figure 2: Custom grip element and fingertip positions

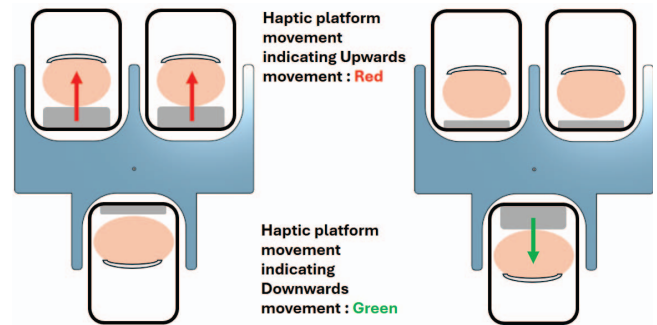


Figure 3: Haptic platform movement for desired directional sensation.

For this study, only the cutaneous force feedback functionality was used. The devices apply pressure to the fingertip through a mobile platform in contact with the fingertip Fig. 2. To simulate the realistic sensation of a "push" by a grasped object, this study explores the use of a forced grip position to generate forces at the correct angles, thereby indicating direction.

To connect the wearable thimbles to a grounded haptic device a custom grip element has been designed that forces the user in the correct position as seen in Fig. 2. The users index and middle finger need to be 180 degrees rotated respect to the thumb, and these need to be kept horizontal respect to the floor; in this way when pressure is applied to the upper side (index and middle fingers) the "push" sensation simulates an upwards direction, and vice versa for the thumb.

This forced position grip allows for the force to be applied to the fingertips perpendicular while having the index and middle finger in the opposite direction to the thumb. This was then installed on a Haply Inverse 3 device that was utilized as positional input for the system. The handle was installed on the end effector of the device, and the user was grasping it while wearing the WEART Touch Diver. To ensure a good clasping force and touch sensation, the TouchDiver end effectors are engaged at the 50 percent of the maximum force strength the devices are able to provide.

To generate the simulated cutaneous directional sensation, the thimble device provides an increase in force to the side the object is simulating movement towards, and slightly decreasing the force provided by the haptic platform on the fingertip at the opposite side. Fig. 3 shows an example of the grip trying to exert the forces for an upwards "push" sensation.

While the device is not providing any feedback, all the actuators are slightly pressuring the users fingertips. This helps with gripping the device, and makes the change in pressure more noticeable once the system starts providing feedback. When the system provides feedback, there are two different modes of operation: a) static

feedback, where the device provides the same force while feedback is required; b) variable feedback, when the system is engaged the force magnitude that the device provides fluctuates based on a wave function.

5 PILOT STUDY

The aim of this study is to assess the performance of a novel interaction system integrating a set of cutaneous thimbles into a desktop interface to control a virtual scenario, and gathering subjective feedback from participants on the realism of the simulated "push" sensation. The focus is on the role of haptic feedback in enhancing user performance within a virtual environment, particularly through cutaneous cues. By using the cutaneous thimbles, participants navigated a virtual workspace to examine how different force patterns impacted their ability to perform a target finding task. The study specifically investigates whether static force application in force cues, such as intensity and frequency compared to static force, affect this particular task performance, including completion time and accuracy.

5.1 Experimental design

The experiment consisted of eight participants, all of them were subjected to all the possible feedback implementations. All participants gave written informed consent before participation. The experiments were approved by the Research Ethics Committee of Aarhus University.

The experiment setup is shown in Fig. 1, where participants interacted with the virtual environment using a Haply Inverse 3 haptic device, with their non-dominant hand while wearing a WEART TouchDIVER device on the index, middle finger, and thumb. The force is provided by the WEART TouchDIVER, and the force feedback was disabled for the Haply inverse 3 used only to track the user's hand pose. Participants were in a seated position while doing the test, they wore a noise-canceling headset with background white noise to minimize external auditory distractions. A monitor in front of the user showed the user interface, where they could see their position (a blue sphere) and the axis of movement (vertical white line). The task for the participant was to control a blue sphere using their hand through the Haply device to reach a series of random invisible targets on a vertical white line indicator on the screen. The seven force variations are used to guide towards the current target. The cutaneous cues applied simultaneous pressure on the index and middle fingers suggesting the user the feeling of upward hand movement, while pressure on the thumb indicates downward movement.

The experiment has been set to test how different force patterns described in table 1 affect the user in three groups: Group 1 is a static force with no variations, group 2 (force patterns 7,8,9) focuses on amplitude modulation, while group 3 (force patterns 10,11,12) focuses on frequency modulation with a control case with no modulation.

The complete task consists of reaching 18 different targets, that were divided into 3 groups and the total travel distance for each group of 6 targets was designed to be constant. The different force patterns tested in the study are summarized in Table 1. The users received force feedback based on which feedback case they were experiencing, this feedback is applied as an increase in pressure to the fingertips via the haptic platforms, this force magnitude has a value range from 0 (minimum) to 1 (maximum) this represents a cutaneous force range from 0 N to 5 N respectively, and can be controlled dynamically. The different cases and their differences are represented in Table 1.

5.2 Evaluation criteria

To evaluate the users performance two key metrics were used, completion time to reach desired target, and distance to desired target

Case	Force Type	Frequency (Hz)	Amplitude
1 (Control)	Static Force	-	0.75
7	Oscillating Force	10	0.1
8	Oscillating Force	10	0.2
9	Oscillating Force	10	0.3
10	Oscillating Force	5	0.2
11	Oscillating Force	10	0.2
12	Oscillating Force	15	0.2

Table 1: Summary of cases with force types, frequencies, and amplitudes.

(accuracy error). These two key characteristics are most common when dealing with input and feedback systems to determine if a feedback system is successful.

The users responded via a questionnaire if any of the patters would feel like a realistic "push" sensation; this evaluation is provided by the participants after experiencing all the feedback modalities. To avoid confusion, participants were asked to respond on how realistic was the feedback compared to a real pushing sensation from a vertically moving object, how realistic was the sensation of "push" for the user.

5.3 Protocol

The experiment begins with a familiarization session where the environment is introduced to the subject and several force patterns were shown to ensure the user could notice and distinguish between the pair of two cues (cutaneous feedback to suggest for upward and downward hand movements) in each of the cases.

Participants were asked to reach 18 target positions (the same for each user) with the guidance of each of the seven cutaneous feedback; to avoid variability, all users for each round started from a central position. To gather the final measurement, the user pressed a key on the keyboard in front of the user once they believed the target was reached. To allow the participants better differentiate between the different force variations for the subjective questionnaire, they were allowed to take note comments about the experience in between each round with a notepad on the side of the keyboard.

After the physical part of the experiment was concluded the users answered a questionnaire to determine the intuitiveness and realism of the different force feedback implementations.

5.4 Statistical analysis

To examine the effects of the independent variable (force modulation) on the dependent variables (completion time and the positioning error), one-way repeated measures ANOVA was chosen to compare these effects between different modalities. This accounted for each of the participant being subjected to all variations. A p value of 0.05 or lower has been considered for statistical significant difference. To test the normality of the data, the Shapiro-Wilk normality test was utilized, a logarithmic transformation for normalization was used before proceeding with the analysis.

6 RESULTS

In terms of completion time no significant effect of the different feedback variations was found. Specifically, the analysis showed that the feedback factor within subjects did not produce a statistically significant main effect, ($F(6, 810) = 1.145, p = 0.334$). This suggests that variations in force feedback levels do not significantly affect participants' completion time.

When interpreting the results descriptively, it can be seen that feedback 1 (control case) has a slightly slower completion time compared to all other feedback modalities that included a force variation Fig. 4.

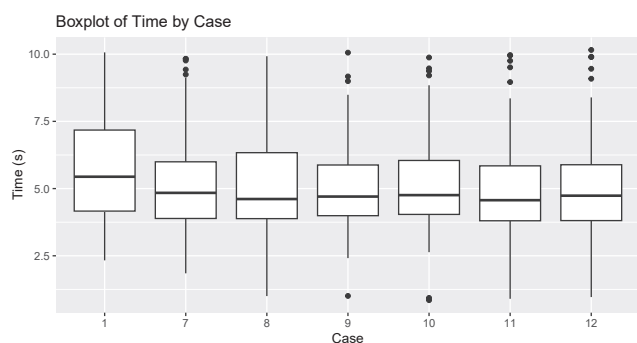


Figure 4: Completion time for each different feedback variation in seconds (lower is better).

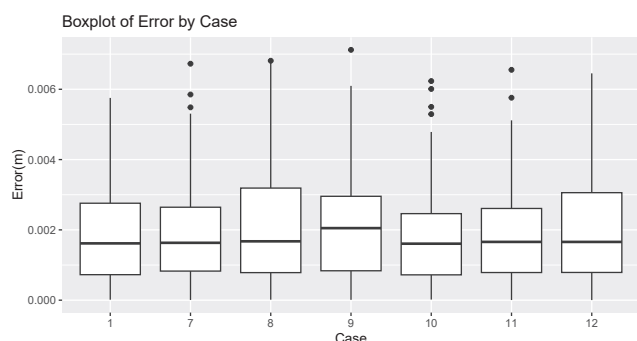


Figure 5: Distance from user final position to target objective for each feedback variation in meters (Lower is better).

A separate one-way repeated measures ANOVA was performed to assess the effect of the provided case on the final position accuracy. The results show no significant effect of feedback case on the error, $F(6,810)=0.967, p=0.446$. This indicates that feedback levels did not significantly influence the error rates in this experiment.

When looking at the results descriptively there is no significant difference between the static force and the feedback modality with force variation Fig. 5. This also indicates that the control group was able to perform without any effect on the outcome despite the feedback being constant.

6.1 Subjective questions results

To evaluate the users experience of the feedback provided to the users, the participants answered a subjective questionnaire about the system. The questions indicated in Table 2 were asked to be answered in a scale from 0 to 100 in 5 degree increments, where 0 is lowest and 100 is highest.

The overall questionnaire results indicate that there is not a large difference between the control case and the other cases, in term of intuitiveness suggesting that the cutaneous force variation did not affect the intuitiveness. Q1 had similar results in most cases, indicating an overall intuitiveness of the feedback implementation. The Q2 aims to answer to find if the system provides a realistic directional "push" sensation; this was asked to be compared to imagining to hold an object moving towards an upwards or downwards direction. The control case produced the highest realism score Fig. 6, followed by case 7, while the other cases vary, this indicates that there is no difference between modulating frequency or amplitude in terms of realism of the feedback.

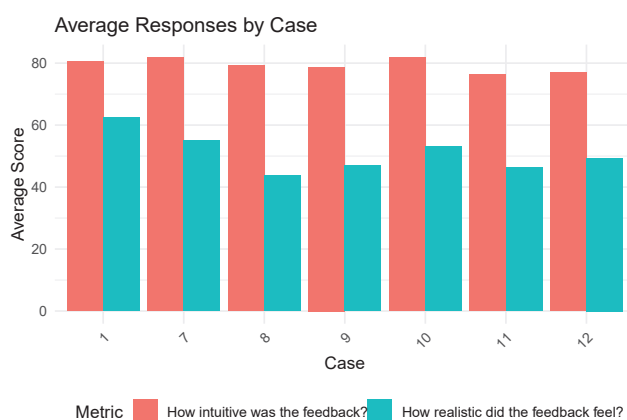


Figure 6: Subjective questionnaire results by case and question (Higher is better).

Question	Description
Q1	How intuitive was the feedback?
Q2	How realistic was the feedback?

Table 2: Questions used for subjective user evaluation.

7 DISCUSSION

Overall the study shows that the different feedback implementations tested have minimal impact on task performance and user experience. Completion time and error rates showed no statistically significant differences across feedback conditions, indicating that force variation does not affect these metrics. Similarly, the questionnaire results revealed no substantial differences in intuitiveness, and while the control case had a higher average score for realism of the feedback, no clear benefit was observed from modulating frequency or amplitude. Overall, the system provides consistent performance and user experience across feedback variations.

This study highlights that static force feedback may be of support for many applications, offering a simple and cost-effective way to provide directional haptic feedback. However, it is important to note that the results of our analysis did not achieve statistical significance. This suggests that while there are trends pointing to the potential utility of static force feedback, further research with larger sample sizes or alternative methodologies is required to confirm these findings. Despite the lack of statistical significance, static force feedback could still be a viable option for applications where simplicity and cost-effectiveness are critical considerations. These findings are relevant for fields like virtual reality and robotics, where the addition of haptic feedback enhances user experience.

8 CONCLUSION

The experimental findings suggest that while no significant results were found, there are clear trends present. Descriptive trends indicated a slight increase in completion time for the feedback modality without force variation. Error rates did not show evident differences in between static force and force variation feedback modalities. Despite the absence of statistically significant effects, this work provides important insights into the design and evaluation of feedback systems. Notably, that static varying the force feedback magnitude does not provide a clear advantage over a simpler implementation of static force. Moreover, this study establishes a foundational understanding of the interplay between feedback modalities and task performance, which can guide future research on optimizing haptic

feedback for interactive systems.

The subjective questionnaire results show minimal differences in intuitiveness between the control and other cases, suggesting that force variation does not affect the intuitiveness of the feedback in this implementation, but the results do show that all of them had a higher level of intuitiveness indicating that the proposed interaction provides a clear feedback to the user. The realism of the interaction was highest in the control case, with case 7 following, indicating no clear advantage in modulating frequency or amplitude towards making the feedback more realistic.

9 FUTURE WORK

This force implementation could be tested in a scenario holding a device that is tethered to the ground versus ungrounded as well as comparing it to other feedback systems in particular tasks, path guidance, object tracking as examples.

Further research could explore whether force-feedback variations provide greater benefits in tasks requiring precision, prolonged use, or heightened immersion. As well as different implementations of this system where the magnitude of the forces has been reduced, one of the limitations presented by testing it with a single type of cutaneous device. Testing with other devices or a custom made device could prove valuable for further evaluations and comparison.

ACKNOWLEDGMENTS

This research was supported by the SENSIBLE project, Symbiotic tEleoperation for Safe and cost-efficient wInd turbine bLade maintenance, which played a pivotal role in enabling the work presented in this work. We gratefully acknowledge the resources provided by AU, which contributed significantly to this outcome.

REFERENCES

- [1] A. Abbasimoshaei, J. Reisinger, C. Neupert, C. Hatzfeld, and W. Zhou. Evaluation of Haptic Systems. In T. A. Kern, C. Hatzfeld, and A. Abbasimoshaei, eds., *Engineering Haptic Devices*, pp. 587–623. Springer International Publishing, Cham, 2023. doi: 10.1007/978-3-031-04536-3_13 2
- [2] P. Ballin. Tactile Design Principles. In *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction*, TEI '21, pp. 1–5. Association for Computing Machinery, New York, NY, USA, Feb. 2021. doi: 10.1145/3430524.3443688 1
- [3] O. N. Bandiaky, S. Lopez, L. Hamon, R. Clouet, A. Soueidan, and L. Le Guehennec. Impact of haptic simulators in preclinical dental education: A systematic review. *Journal of Dental Education*, 88(3):366–379, 2024. .eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/jdd.13426>. doi: 10.1002/jdd.13426 1
- [4] L. Becker, B. Weber, and N. Bechtel. Haptic Guidance for Teleoperation: Optimizing Performance and User Experience. In H. Seifi, A. M. L. Kappers, O. Schneider, K. Drewing, C. Pacchierotti, A. Abbasimoshaei, G. Huisman, and T. A. Kern, eds., *Haptics: Science, Technology, Applications*, pp. 129–137. Springer International Publishing, Cham, 2022. doi: 10.1007/978-3-031-06249-0_15 1
- [5] S. Choi and K. J. Kuchenbecker. Vibrotactile Display: Perception, Technology, and Applications. *Proceedings of the IEEE*, 101(9):2093–2104, Sept. 2013. Conference Name: Proceedings of the IEEE. doi: 10.1109/JPROC.2012.2221071 1
- [6] H. Culbertson, S. B. Schorr, and A. M. Okamura. Haptics: The Present and Future of Artificial Touch Sensation. *Annual Review of Control, Robotics, and Autonomous Systems*, 1(Volume 1, 2018):385–409, May 2018. Publisher: Annual Reviews. doi: 10.1146/annurev-control-060117-105043 1
- [7] A. Gani, O. Pickering, C. Ellis, O. Sabri, and P. Pucher. Impact of haptic feedback on surgical training outcomes: A Randomised Controlled Trial of haptic versus non-haptic immersive virtual reality training. *Annals of Medicine and Surgery*, 83:104734, Nov. 2022. doi: 10.1016/j.amsu.2022.104734 1
- [8] G. J. Gerling. SA-I Mechanoreceptor Position in Fingertip Skin May Impact Sensitivity to Edge Stimuli. *Applied Bionics and Biomechanics*, 7(1):874936, 2010. .eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1080/11762320903069992>. doi: 10.1080/11762320903069992 1
- [9] A. Madani, P. P. Niaz, B. Guler, Y. Aydin, and C. Basdogan. Robot-Assisted Drilling on Curved Surfaces with Haptic Guidance under Adaptive Admittance Control. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 3723–3730, Oct. 2022. ISSN: 2153-0866. doi: 10.1109/IROS47612.2022.9982000 1
- [10] J. E. Piercy and A. J. Brammer. Development of vibrotactile measurement techniques for assessing mechanoreceptor performance at the fingertip. *The Journal of the Acoustical Society of America*, 80(S1):S40, Aug. 2005. doi: 10.1121/1.2023787 1
- [11] D. Prattichizzo, F. Chinello, C. Pacchierotti, and M. Malvezzi. Towards Wearability in Fingertip Haptics: A 3-DoF Wearable Device for Cutaneous Force Feedback. *IEEE Transactions on Haptics*, 6(4):506–516, Oct. 2013. Conference Name: IEEE Transactions on Haptics. doi: 10.1109/TOH.2013.53 1
- [12] U. Radhakrishnan, L. Kuang, K. Koumaditis, F. Chinello, and C. Pacchierotti. Haptic Feedback, Performance and Arousal: A Comparison Study in an Immersive VR Motor Skill Training Task. *IEEE Transactions on Haptics*, 17(2):249–262, Apr. 2024. Conference Name: IEEE Transactions on Haptics. doi: 10.1109/TOH.2023.3319034 1, 2
- [13] C. Shi, J. Madera, H. Boyea, and A. M. Fey. Haptic Guidance Using a Transformer-Based Surgeon-Side Trajectory Prediction Algorithm for Robot-Assisted Surgical Training. In *2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, pp. 1942–1949, Aug. 2023. ISSN: 1944-9437. doi: 10.1109/RO-MAN57019.2023.10309357 1
- [14] J. L. Sullivan, S. Pandey, M. D. Byrne, and M. K. O'Malley. Haptic Feedback Based on Movement Smoothness Improves Performance in a Perceptual-Motor Task. *IEEE Transactions on Haptics*, 15(2):382–391, Apr. 2022. Conference Name: IEEE Transactions on Haptics. doi: 10.1109/TOH.2021.3126017 2
- [15] D. Tsetserukou, K. Sato, A. Neviarouskaya, N. Kawakami, and S. Tachi. FlexTorque: innovative haptic interface for realistic physical interaction in virtual reality. In *ACM SIGGRAPH ASIA 2009 Sketches*, SIGGRAPH ASIA '09, p. 1. Association for Computing Machinery, New York, NY, USA, 2009. doi: 10.1145/1667146.1667159 1
- [16] C.-M. Wu and S. Smith. A haptic keypad design with a novel interactive haptic feedback method. *Journal of Engineering Design*, 26(4-6):169–186, June 2015. Publisher: Taylor & Francis .eprint: <https://doi.org/10.1080/09544828.2015.1030372>. doi: 10.1080/09544828.2015.1030372 1